

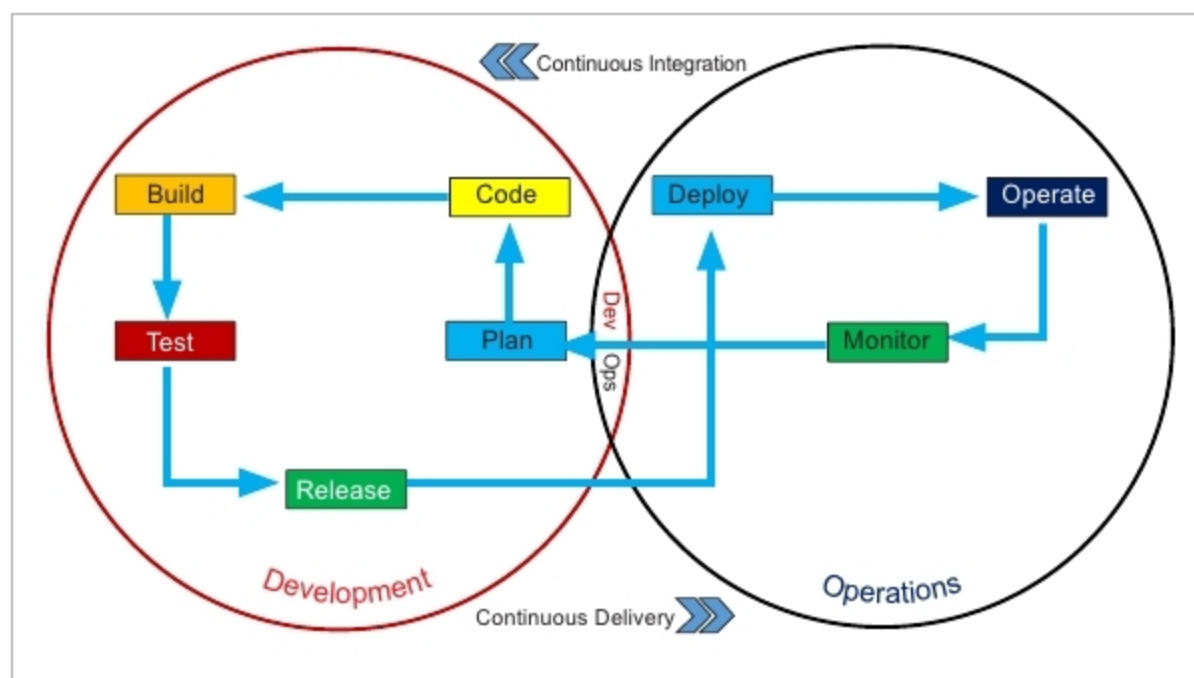


Figure 1: DevOps life cycle

DevOps life cycle and tools

The DevOps life cycle consists of eight stages – plan, code, build, test, release, deploy, operate and monitor. There are various free and open source software (FOSS) tools that can be used to improve and automate the DevOps life cycle.

The eight stages of DevOps are explained below.

1. **Plan:** This is the starting stage of DevOps where the planning for the software development is done by the development team. Everything from the software requirements to the development timeline is planned and the team works accordingly. FOSS tools like Redmine, Trac and Git are used at this stage.
2. **Code:** After everything is planned, the coding for the software development is done. Coding can be done from scratch or can be reused, depending on the requirements. Many FOSS tools can be used for coding. Git is one such tool that is used to automate the process.
3. **Build:** After the coding is done, it is shared with other software engineers of the development team. The code is either approved or rejected after it is reviewed. If it's approved, then it is merged with the main codebase of the repository. FOSS tools like Gradle can be used to automate this process.
4. **Test:** Once the new code is merged with the codebase, it is tested in a virtual environment, using a VM or Kubernetes. A series of both manual and automated tests are conducted at this stage. This is the most crucial stage of DevOps and should go through without failure. This stage could also cause bottlenecks and increase the timeline. In order to complete the stage successfully without much delay, various continuous testing tools are used. FOSS tools like CruiseControl and Selenium are used to automate the process.
5. **Release:** After the code is tested successfully, it is prepared for release. At this stage the development team decides which features of the software product should be enabled or disabled by default and when it should be released. This is the final stage for the DevOps development cycle, and after this the software/code is delivered to the operations team. This stage can also be automated using FOSS tools like Jenkins and Bamboo.
6. **Deploy:** This stage belongs to the operations team and it starts after the release of the software by the development team. The software is then deployed by the operations team using various tools. The release is configured according to the

requirements of the operations team. FOSS tools like Puppet, Ansible and Saltstack are used to automate the deployment process.

7. **Operate:** After the software has been updated and configured, the operations team starts operating the products and services with the updated software. This is done by either using proprietary or FOSS tools.
8. **Monitor:** This is the final stage for the IT operations team and also for the DevOps life cycle. Here the customer requirements are gathered and the data is sent to the development team to update the software product/service for the next iteration of DevOps. FOSS tools like Nagios are used by the operations team to automate the monitoring process.

Benefits of DevOps

DevOps has many more benefits than the agile software development process. In case of the latter, the development process progresses through four main stages – planning, coding, testing and release. The planning for the software development is done based on customer requirements and involves assigning various tasks to the stakeholders, like designing flowcharts, writing documentation, estimating timelines, etc. After the planning is done, the coding for the software development starts and a working prototype is created. The prototype is then tested in the third (testing) stage. The testing stage is the most important part of the agile software development life cycle because it can affect the overall quality of the software. This stage can also cause bottlenecks because the time required for software testing cannot be estimated precisely and could go beyond the project's timeline.

After successfully testing the software, it is then released to the customer in the fourth stage. However, if the customer finds any problems with the software, then the development team takes the customer's feedback for another iteration of the development